

Week 16 Office Hour Notes

16.0 Errata

16.1 The Lowenheim-Skolem Theorems

Theorem 16.1.1. (Lowenheim Skolem Theorems) Let Σ be theory of wff's satisfied by a structure, and let X be an infinite set with $|X| \geq |\mathcal{L}|$. Then there exists a structure $\mathfrak{B}$ elementary to $\mathfrak{A}$ (and hence also satisfying $\mathfrak{A}$) such that $|\mathfrak{B}| = |X|$.

Outline of the proof. We have the following cases

Case 1. (The Downward Lowenheim Skolem theorem) Assume $|X| = |\mathcal{L}|$. Then we use the proof of the Completeness Theorem to construct a model $\mathfrak{B}$ of $\text{Th}_{\mathcal{L}}(\mathfrak{A})$ that is elementary equivalent to $\mathfrak{A}$ and to where we have $|\mathfrak{B}| = |\mathcal{L}|$. Details will be worked out in a later draft.

Case 2. (The Upward Lowenheim Skolem theorem) Assume that $|X| > |\mathcal{L}|$. Since by Case 1 we can construct a structure $\mathfrak{A}'$ such that $|\mathfrak{A}'| = |\mathcal{L}|$ that is elementarily equivalent to $\mathfrak{A}$, assume without loss of generality that $|\mathfrak{A}| = |\mathcal{L}|$.

Then add constant symbols $C := \{c_x\}_{x \in X}$ to the language $\mathcal{L}' := \mathcal{L} \cup C$. Since $|X| > |\mathcal{L}|$, we find that

$|\mathcal{L}'| = |\mathcal{L} \sqcup X| = \max(|\mathcal{L}|, |X|) = |X|$. (Details of this part of this proof will be worked out in the upcoming [Optional Office Hours Part II Section II.5](#))

Then let

$$\Sigma := \{\neg c_x = c_y : x \neq y \in X\}.$$

We next want to show that $\Gamma := \Sigma \cup \text{Th}_{\mathcal{L}}(\mathfrak{A})$ is satisfiable. By compactness, it suffices to show that it's finitely satisfiable. Given finite $\Gamma_0 \subset \Gamma$, note that $\mathfrak{A}_0$ where $|\mathfrak{A}_0| := |\mathfrak{A}|$, and all the functions and relations of $\mathfrak{A}_0$, corresponding to the relation symbols in $\mathcal{L}$, agree with $\mathfrak{A}$, and for constant symbols C , we note that since $|\mathfrak{A}|$ is infinite (since WLOG, we assumed $|\mathfrak{A}| = |\mathcal{L}|$), for each $c_1, \dots, c_N \in C$ mentioned in Γ_0 , we can choose distinct $a_1, \dots, a_N, a_{N+1} \in |\mathfrak{A}|$ and set

$$(c_k)^{\mathfrak{A}_0} := a_k \quad (1 \leq k \leq N),$$

$(c^{\mathfrak{U}_0}) := a_{N+1}$ for all $c \in C \setminus \{c_1, \dots, c_N\}$.

Then it follows that $\mathfrak{U}_0$ satisfies Γ_0 since $\mathfrak{U}_0$ (as an extension of $\mathfrak{U}$) satisfies everything in $\Gamma_0 \cap \text{Th}_{\mathcal{L}}(\mathfrak{U})$, and since $\{a_1, \dots, a_N\}$ were chosen to be distinct, $\mathfrak{U}_0$ also satisfies everything in $\Gamma_0 \cap \Sigma$. This completes the argument that Γ is finitely satisfiable.

Since Γ is satisfiable, we can choose a model $\mathfrak{B}$ of Γ , and since using **Case 1** we can choose a model $\mathfrak{B}'$ elementarily equivalent to $\mathfrak{B}$ such that $|\mathfrak{B}'| = |\mathcal{L}'| = |X|$, we can assume without loss of generality that $|\mathfrak{B}| = |X|$, and it remains to show that $\mathfrak{U} \equiv \mathfrak{B}$. This is immediate from the fact that $\text{Th}_{\mathcal{L}}(\mathfrak{B}) = \text{Th}_{\mathcal{L}}(\mathfrak{U})$ by construction of $\mathfrak{B}$ as a model of Γ . $\square$

16.2 Complete Theories

Next, we want to talk about theories that are complete (yaaaay another definition of "complete")

Definition 16.2.1. We state that a theory T is **complete** if for every sentence σ either $\sigma \in T$ or $\neg\sigma \in T$ (but not both).

Proposition 16.2.2. (*Basic Properties of Complete Sets*) Let T be a complete theory.

- (i) A complete theory is consistent.
- (ii) A theory is complete if and only if it is maximally satisfiable.
- (iii) If $\mathfrak{U}$ is a model of T , then $\text{Th}(\mathfrak{U}) = T = \text{Th}(T)$.
- (iv) If $T \subset T'$ and T' is consistent, then $T = T'$.

The proofs of the above proposition will be left to the reader, here's a proposition (that I'll state as a lemma since we need it to prove the Vaught test)

Lemma 16.2.3. T is a complete theory if and only if T is satisfiable and any two models are elementarily equivalent.

Proof.

$\implies$ Suppose T is a complete theory. Then given two models $\mathfrak{U}, \mathfrak{B}$, we find by completeness and **Proposition 16.2.2 (iii)** that $\text{Th}(\mathfrak{U}) = T = \text{Th}(\mathfrak{B})$.

$\Leftarrow$ Suppose conversely and contrapositively that T is not a complete theory. If T is unsatisfiable, then we are finished, and we can suppose without loss of generality that T is satisfiable. Then since T is incomplete, we can choose a sentence σ such that $\sigma, \neg\sigma \notin T$. Since $\sigma, \neg\sigma \notin T \implies T \not\vdash \sigma, T \not\vdash \neg\sigma$, (since T is a theory) we find that $T \cup \{\sigma\} \not\vdash \neg\sigma$ and $T \cup \{\neg\sigma\} \not\vdash \sigma$, since otherwise using *Reducto Ad-Adsurdum* we would have either $T \vdash \sigma$ or $T \vdash \neg\sigma$, contradicting the fact that $T \not\vdash \sigma, T \not\vdash \neg\sigma$. It follows that both $T \cup \{\sigma\}$ and $T \cup \{\neg\sigma\}$ are consistent, hence by completeness both $T \cup \{\sigma\}$ and $T \cup \{\neg\sigma\}$ are satisfiable.

Choose models $\mathfrak{A}$ and $\mathfrak{B}$ of $T \cup \{\sigma\}$ and $T \cup \{\neg\sigma\}$, respectively. Since $\models_{\mathfrak{A}} \sigma$, but $\models_{\mathfrak{B}} \neg\sigma \implies \not\models_{\mathfrak{A}} \sigma$, we find $\mathfrak{A}$ and $\mathfrak{B}$ don't satisfy the same formulas; therefore they're not elementarily equivalent. $\square$

in mathematics, we're interested in complete theories, especially those that are computable/decidable, because then they can be proved using computer algorithms.

Here's one way of proving that a theory is complete

Definition 16.2.5. Let T be a theory, let κ be a cardinal (as defined in **Definition II.2.4** of the [Optional Office Hour Notes Part II](#)). We say that T is κ -categorical if all models of cardinality κ are all isomorphic.

Theorem 16.2.6. (The Vaught-Test) Let T be a satisfiable theory with no infinite models. If T is κ -categorical, then T is complete.

Proof. Suppose that T is κ -categorical. By **Lemma 16.2.3**, it shall suffice to show that every model of T is elementarily equivalent. Let $\mathfrak{A}$ and $\mathfrak{B}$ be two models of T . By the Lowenheim-Skolem Theorem, there exists models $\mathfrak{A}'$ and $\mathfrak{B}'$ of T such that $\mathfrak{A} \equiv \mathfrak{A}'$ and $\mathfrak{B} \equiv \mathfrak{B}'$, and

$$|\mathfrak{A}'| = |\mathfrak{B}'| = |\kappa|.$$

Since T is κ -categorical, we find $\mathfrak{A}' \cong \mathfrak{B}'$, implying that $\mathfrak{A}' \equiv \mathfrak{B}'$, and we conclude that

$$\mathfrak{A} \equiv \mathfrak{A}' \equiv \mathfrak{B}' \equiv \mathfrak{B}. \quad \square$$

%consider giving examples of complete theories where the Vaught Test is used and consider other examples mention in that section

Note that mathematicians in the 20th century, as a result of finding the paradoxical nature of naive set theory (see **Section II.3** of the [Optional Office Notes Part II](#))--and in particular a

mathematician named David Hilbert made one of many conjectures of this--sought to find a theory of arithmetic (i.e., a theory that is able to satisfy all the properties of Peano Arithmetic) that is both computable and complete. Unfortunately for those hoping to find a positive answer, the famous (or infamous) mathematician Kurt Godel proved otherwise.

Theorem 16.2.7. (*Godel's First Incompleteness Theorem*) Let T be a consistent theory in the language $\mathcal{L} := (0, S, +, \cdot, <)$ that satisfies the properties of Peano Arithmetic. Then T is incomplete.

My office hour will be 3pm-4:30pm Tuesday Dec. 17th, and we will talk about The Incompleteness Theorem.